

# Online Examination Report

|                |                                |
|----------------|--------------------------------|
| Date:          | 21.07.2014                     |
| Attempts:      | 1                              |
| Examinee:      | Shyaka, Daniel                 |
| Date of birth: | 5/26/1993                      |
| Subject:       | 030-Flight Performance&Planing |
| Result:        | 41,49 %                        |
| Time used:     | 00:00:13 (HH:MM:SS)            |

## Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

## Attachment 1: List of result by question

| CQB Number | Internal Number | Syllabus reference | Chapter                                   | Points | Max. Points |
|------------|-----------------|--------------------|-------------------------------------------|--------|-------------|
| 1          | 375             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 1,00   | 1,00        |
| 2          | 374             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 3          | 373             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 4          | 381             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 5          | 361             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 6          | 379             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 7          | 388             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 1,00   | 1,00        |
| 8          | 391             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 9          | 382             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 10         | 390             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 1,00   | 1,00        |
| 11         | 387             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 12         | 377             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 1,00   | 1,00        |
| 13         | 378             | 31.                | MASS AND BALANCE - AEROPLANES/HELICOPTERS | 0,00   | 1,00        |
| 14         | 257             | 31.3.1             | Definition of centre of gravity           | 1,00   | 1,00        |
| 15         | 278             | 31.3.3             | Basic calculations of CG                  | 0,00   | 2,00        |
| 16         | 410             | 32.                | PERFORMANCE - AEROPLANES                  | 1,00   | 1,00        |
| 17         | 339             | 32.                | PERFORMANCE - AEROPLANES                  | 1,00   | 1,00        |
| 18         | 152             | 32.2.5.4           | Landing                                   | 1,00   | 1,00        |
| 19         | 210             | 32.3.3.1           | Take-off                                  | 1,00   | 1,00        |
| 20         | 46              | 32.3.3.1           | Take-off                                  | 1,00   | 1,00        |
| 21         | 115             | 32.4.1             | Take - off                                | 0,00   | 1,00        |
| 22         | 219             | 32.4.1.7           | Take-off climb                            | 0,00   | 1,00        |
| 23         | 256             | 32.4.1.7           | Take-off climb                            | 1,00   | 1,00        |
| 24         | 119             | 32.4.1.8           | Obstacle-limited take-off                 | 1,00   | 1,00        |
| 25         | 68              | 32.4.3             | Cruise                                    | 1,00   | 1,00        |
| 26         | 129             | 32.4.3             | Cruise                                    | 0,00   | 1,00        |
| 27         | 15              | 32.4.3             | Cruise                                    | 1,00   | 1,00        |
| 28         | 47              | 32.4.3             | Cruise                                    | 1,00   | 1,00        |

| CQB Number       | Internal Number | Syllabus reference | Chapter                                                                | Points       | Max. Points  |
|------------------|-----------------|--------------------|------------------------------------------------------------------------|--------------|--------------|
| 29               | 154             | 32.4.3.4           | Long Range Cruise                                                      | 0,00         | 1,00         |
| 30               | 21              | 32.4.5.1           | Descent techniques                                                     | 0,00         | 1,00         |
| 31               | 421             | 33.                | FLIGHT PLANNING AND FLIGHT MONITORING                                  | 0,00         | 1,00         |
| 32               | 415             | 33.                | FLIGHT PLANNING AND FLIGHT MONITORING                                  | 1,00         | 1,00         |
| 33               | 92              | 33.                | FLIGHT PLANNING AND FLIGHT MONITORING                                  | 0,00         | 1,00         |
| 34               | 416             | 33.                | FLIGHT PLANNING AND FLIGHT MONITORING                                  | 1,00         | 1,00         |
| 35               | 414             | 33.                | FLIGHT PLANNING AND FLIGHT MONITORING                                  | 0,00         | 1,00         |
| 36               | 2               | 33.2.1             | IFR Navigation plan                                                    | 0,00         | 1,00         |
| 37               | 175             | 33.2.1.2           | Courses and distances from en-route charts                             | 0,00         | 1,00         |
| 38               | 122             | 33.3.3             | Specific fuel calculation procedures                                   | 0,00         | 1,00         |
| 39               | 224             | 33.4.2.1           | Extraction and analysis of relevant data from meteorological documents | 0,00         | 1,00         |
| 40               | 30              | 33.5.1.1           | Format of Flight Plan                                                  | 0,00         | 1,00         |
| <b>Total:41%</b> |                 |                    |                                                                        | <b>17,00</b> | <b>41,00</b> |

## Attachment 2: List of result by topic

**Licence**

**Date: 17.07.2014**

**Examinee: Shyaka, Daniel**

**Location: Uganda Civil Aviation Authority**

**Your Result: 17,00 from 41,00 Points (41,46 %)**

| Topics                                                               | Amount of Questions | Points achieved | Maximum Points | Percent      |
|----------------------------------------------------------------------|---------------------|-----------------|----------------|--------------|
| 31. MASS AND BALANCE - AEROPLANES/HELICOPTERS                        | 15                  | 5,00            | 16,00          | 31,25        |
| 31.3. FUNDAMENTALS OF CG CALCULATIONS                                | 2                   | 1,00            | 3,00           | 33,33        |
| 32. PERFORMANCE - AEROPLANES                                         | 15                  | 10,00           | 15,00          | 66,67        |
| 32.2. PERFORMANCE CLASS B - SINGLE-ENGINE AEROPLANES                 | 1                   | 1,00            | 1,00           | 100,00       |
| 32.3. PERFORMANCE CLASS B - MULTI-ENGINE AEROPLANES                  | 2                   | 2,00            | 2,00           | 100,00       |
| 32.4. PERFORMANCE CLASS A - AEROPLANES CERTIFICATED UNDER CS 25 ONLY | 10                  | 5,00            | 10,00          | 50,00        |
| 33. FLIGHT PLANNING AND FLIGHT MONITORING                            | 10                  | 2,00            | 10,00          | 20,00        |
| 33.2. FLIGHT PLANNING FOR IFR FLIGHTS                                | 2                   | 0,00            | 2,00           | 0,00         |
| 33.3. FUEL PLANNING                                                  | 1                   | 0,00            | 1,00           | 0,00         |
| 33.4. PRE-FLIGHT PREPARATION                                         | 1                   | 0,00            | 1,00           | 0,00         |
| 33.5. ICAO FLIGHT PLAN (ATS Flight Plan)                             | 1                   | 0,00            | 1,00           | 0,00         |
| <b>Total</b>                                                         | <b>40</b>           | <b>17,00</b>    | <b>41,00</b>   | <b>41,46</b> |

Candidate: Daniel Shyaka  
Date of birth: 1993-05-26

Subject: 030-Flight Performance&Planing  
Date: 2014-07-17



1

What is the maximum weight that could be added at Station 150.0 without exceeding the aft CG limit?

Aircraft weight 5,000 lb  
CG location Station 80.0

☐ 69.5 pounds.

☐ 70.0 pounds.

☒ 35.9 pounds.

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2

**How much weight could be added at Station 120 without exceeding the aft CG limit?**

**Aircraft weight 9,500 lb**

**CG location Station 90.0**

**Aft CG limit Station 90.5**

☒ **61.0 pounds.**

☐ **161.0 pounds.**

☐ **110.5 pounds.**

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3

**How much weight could be added at Station 160 without exceeding the aft CG limit?**

Aircraft weight 8,300 lb  
CG location Station 90.0  
Aft CG limit Station 90.5

☒ 13.9 pounds.

☐ 16.5 pounds.

☐ 59.7 pounds.

4

Could 100 pounds of weight be shifted from Station 30.0 to Station 120.0 without exceeding the aft CG limit?

Total weight 4,750 lb

CG location Station 115.8

☐ No; the new CG would be located at Station 118.15.

☐ Yes; the new CG would be located at Station 117.69.

☒ Yes; the CG would remain at Station 115.8.



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5

What is the maximum allowable weight that may be carried on a pallet which has the dimensions of 24.6 x 68.7 inches?  
Floor load limit - 85 lb/sq ft  
Pallet weight - 44 lb  
Tiedown devices - 29 lb

☒ 968.6 pounds.

☐ 953.6 pounds.

☐ 924.5 pounds.

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6

How much weight must be shifted from Station 150.0 to Station 30.0 to move the CG to exactly the aft CG limit?

Total weight 7,500 lb

CG location Station 80.5

☐ 68.9 pounds.

☐ 62.5 pounds.

☒ 65.8 pounds.

7

**What is characteristic of the indicated airspeed if the CG is at the most forward allowable position and constant power and altitude are maintained?**

- ☐ Indicated airspeed will be greater than it would be with the CG in the most rearward allowable position.
- ☒ Indicated airspeed will be less than it would be with the CG in the most rearward allowable position.
- ☐ There is no relationship between CG location and indicated airspeed.

8

**As the CG location is changed, recovery from a stall becomes progressively**

- ☐ more difficult as the CG moves rearward.
- ☒ more difficult as the CG moves either forward or rearward.
- ☐ less difficult as the CG moves rearward.

9

The center of gravity of an aircraft can be determined by

- ☒ dividing total arm by total moments.
- ☐ multiplying total arm by total weight.
- ☐ dividing total moment by total weight.

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10

Under which condition is a forward CG most critical?

☒ On landing.

☐ On takeoff.

☐ When in an unusual attitude.

11

**As the CG moves aft, an aircraft becomes**

- ☐ less stable and less controllable.
- ☒ more stable and controllable as long as the aft CG is not exceeded.
- ☐ less stable, yet easier to control.

12

What is the location of the CG if 146 pounds are removed from Station 150?

Aircraft weight 7,152 lb

CG location Station 82

☐ 83.4.

☒ 80.6.

☐ 81.3.



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13

**What is the location of the CG if 60 pounds are removed from Station 70?**

**Aircraft weight 8,420 lb**

**CG location Station 85**

☐ 84.1.

☒ 84.9.

☐ 85.1.

14

When an aircraft is stationary on the ground, its total weight will act vertically

- ☐ *through its centre of pressure.*
- ☐ *through a point defined as the datum point.*
- ☐ *through the main wheels of its undercarriage assembly.*
- ☒ *through its centre of gravity.*

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15

(Please view annex 031-038 rev. 05.11.2004)  
For the following boom to be in balance:

☒  $A = B (Fa + Fb)$

☐  $A = B \times Fb / Fa$

☐  $A = B \times Fa / Fb$

☐  $A = B + Fb / Fa$

See Attachments:

1. 031-038.gif

Annex

(Attachment  
No.1)

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16

At what minimum speed will dynamic hydroplaning begin if a tire has an air pressure of 70 PSI?

☐ 80 knots.

☐ 85 knots.

☒ 75 knots.

17

**(Refer to Figure 89.) Which of the following configurations will result in the shortest landing distance over a 50-foot obstacle to a wet runway?**

See Attachments:

1. atp\_89.jpg

figure 89

(Attachment  
No.2)

- ☐ Brakes and spoilers at 122,500 pounds gross weight.
- ☒ Brakes, spoilers, and reversers at 131,000 pounds gross weight.
- ☐ Brakes and reversers at 124,000 pounds gross weight.

18

Which of the following factors will lead to an increase of ground distance during a glide, while maintaining the appropriate minimum glide angle speed?

- ☒ Tailwind.
- ☐ Decrease of aircraft mass.
- ☐ Headwind.
- ☐ Increase of aircraft mass.

19

**Due to standing water on the runway the field length limited take-off mass will be**

☐ higher.

☒ lower.

☐ only higher for three and four engine aeroplanes.

☐ unaffected.

20

**An increase in atmospheric pressure has, among other things, the following consequences on take-off performance:**

- ☒ **a reduced take-off distance and improved initial climb performance**
- ☐ **an increased take-off distance and improved initial climb performance**
- ☐ **a reduced take-off distance and degraded initial climb performance**
- ☐ **an increases take-off distance and degraded initial climb performance**



21

**When the outside air temperature increases, then**

- ☐ the field length limited take-off mass and the climb limited take-off mass increases.
- ☒ the field length limited take-off mass decreases but the climb limited take-off mass increases.
- ☐ the field length limited take-off mass and the climb limited take-off mass decreases.
- ☐ the field length limited take-off mass increases but the climb limited take-off mass decreases.

22

**During climb to the cruising level, a headwind component**

- ☐ decreases the ground distance flown during that climb.
- ☐ increases the climb time.
- ☒ decreases the climb time.
- ☐ increases the amount of fuel for the climb.

23

Which statement is correct?

- ☒ *The climb limited take-off mass depends on pressure altitude and outer air temperature*
- ☐ *The climb limited take-off mass will increase if the headwind component increases.*
- ☐ *The climb limited take-off mass increases when a larger take-off flap setting is used.*
- ☐ *The performance limited take-off mass is the highest of:  
field length limited take-off mass  
climb limited take-off mass  
obstacle limited take-off mass.*

24

Regarding the obstacle limited take-off mass, which of the following statements is correct?

- ☐ The maximum bank angle which can be used is 10°.
- ☐ Wind speed plays no role when calculating this particular mass.
- ☒ A take-off in the direction of an obstacle is also permitted in tail wind condition.
- ☐ The obstacle limited mass can never be lower than the climb limited take-off mass.

25

**At constant thrust and constant altitude the fuel flow of a jet engine**

- ☒ increases slightly with increasing airspeed.
- ☐ decreases slightly with increasing airspeed.
- ☐ increases with decreasing OAT.
- ☐ is independent of the airspeed.

26

**A constant headwind component**

- ☐ decreases the angle of climb.
- ☒ increases the best rate of climb.
- ☐ increases the angle of flight path during climb.
- ☐ increases the maximum endurance.

27

The maximum horizontal speed occurs when:

- ☐ The thrust is equal to minimum drag.
- ☒ The maximum thrust is equal to the total drag.
- ☐ The thrust is equal to the maximum drag.
- ☐ The thrust does not increase further with increasing speed.

28

**The optimum cruise altitude is**

- ☒ the pressure altitude at which the best specific range can be achieved.
- ☐ the pressure altitude at which the speed for high speed buffet as TAS is a maximum.
- ☐ the pressure altitude up to which a cabin altitude of 8000 ft can be maintained.
- ☐ the pressure altitude at which the fuel flow is a maximum.



29

**A lower airspeed at constant mass and altitude requires**

- ☐ less thrust and a lower coefficient of lift.
- ☒ more thrust and a lower coefficient of drag.
- ☐ a higher coefficient of lift.
- ☐ more thrust and a lower coefficient of lift.

30

Which statement is correct for a descent without engine thrust at maximum lift to drag ratio speed?

- ☐ A tailwind component increases fuel and time to descent.
- ☐ A tailwind component increases the ground distance.
- ☐ A headwind component increases the ground distance.
- ☒ A tailwind component decreases the ground distance.

31

**How is your flight plan closed when your destination airport has IFR conditions and there is no control tower or Flight control Center(FCC) on the field?**

- ☒ You may close your flight plan any time after starting the approach by contacting any FCC or ATC facility.
- ☐ Upon landing, you must close your flight plan by radio or by telephone to any FCC or ATC facility.
- ☐ The ARTCC controller will close your flight plan when you report the runway in sight.

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32

If fuel consumption is 15.3 gallons per hour and groundspeed is 167 knots, how much fuel is required for an aircraft to travel 620 NM?

☐ 60 gallons.

☐ 63 gallons.

☒ 57 gallons.

33

From which of the following would you expect to find details of the Search and Rescue organisation and procedures (SAR) ?

- ☐ NOTAM
- ☐ ATCC broadcasts
- ☐ AIP (Aeronautical Information Publication)
- ☒ SIGMET

34

If an aircraft is consuming 91 pounds of fuel per hour and groundspeed is 168 knots, how much fuel is required to travel 457 NM?

☐ 265 pounds.

☒ 248 pounds.

☐ 291 pounds.

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35

**GIVEN:**

Usable fuel at takeoff 40 gal  
Fuel consumption rate 12.2 gal/hr  
Constant groundspeed 120 kts  
Flight time since takeoff 1 hr 30 min

☒ 215 NM.

☐ 121 NM.

☐ 176 NM.

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36

In an ATS flight plan, Item 15 (route), a cruising pressure altitude of 32000 feet would be entered as :

☐ D F320

☐ A FL320

☒ C 32000

☐ B S3200



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37

You are flying a constant compass heading of  $252^\circ$ . Variation is  $22^\circ\text{E}$ , deviation is  $3^\circ\text{W}$  and your INS is showing a drift of  $9^\circ$  right. True track is ?

☐  $224^\circ$

☐  $280^\circ$

☐  $262^\circ$

☒  $242^\circ$

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38

An aircraft flies at an airspeed of 380 kt. It flies from A to B and back to A. Distance AB = 480 NM. When going from A to B, it experiences a headwind component = 60 kt. The wind remains constant. The duration of the flight will be:

☒ 2h 10min

☐ 2h 32min

☐ 3h 00min

☐ 2h 35min

39

A METAR reads : SA1430 35002KT 7000 SKC 21/03 Q1024 =  
Which of the following information is contained in this METAR?

- ☐ *runway in use*
- ☒ *period of validity*
- ☐ *temperature/dewpoint*
- ☐ *day/month*

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40

If your destination airport has no ICAO indicator, in the appropriate box of your flight plan, you write:

☒ C ///

☐ B XXXX

☐ A AAAA

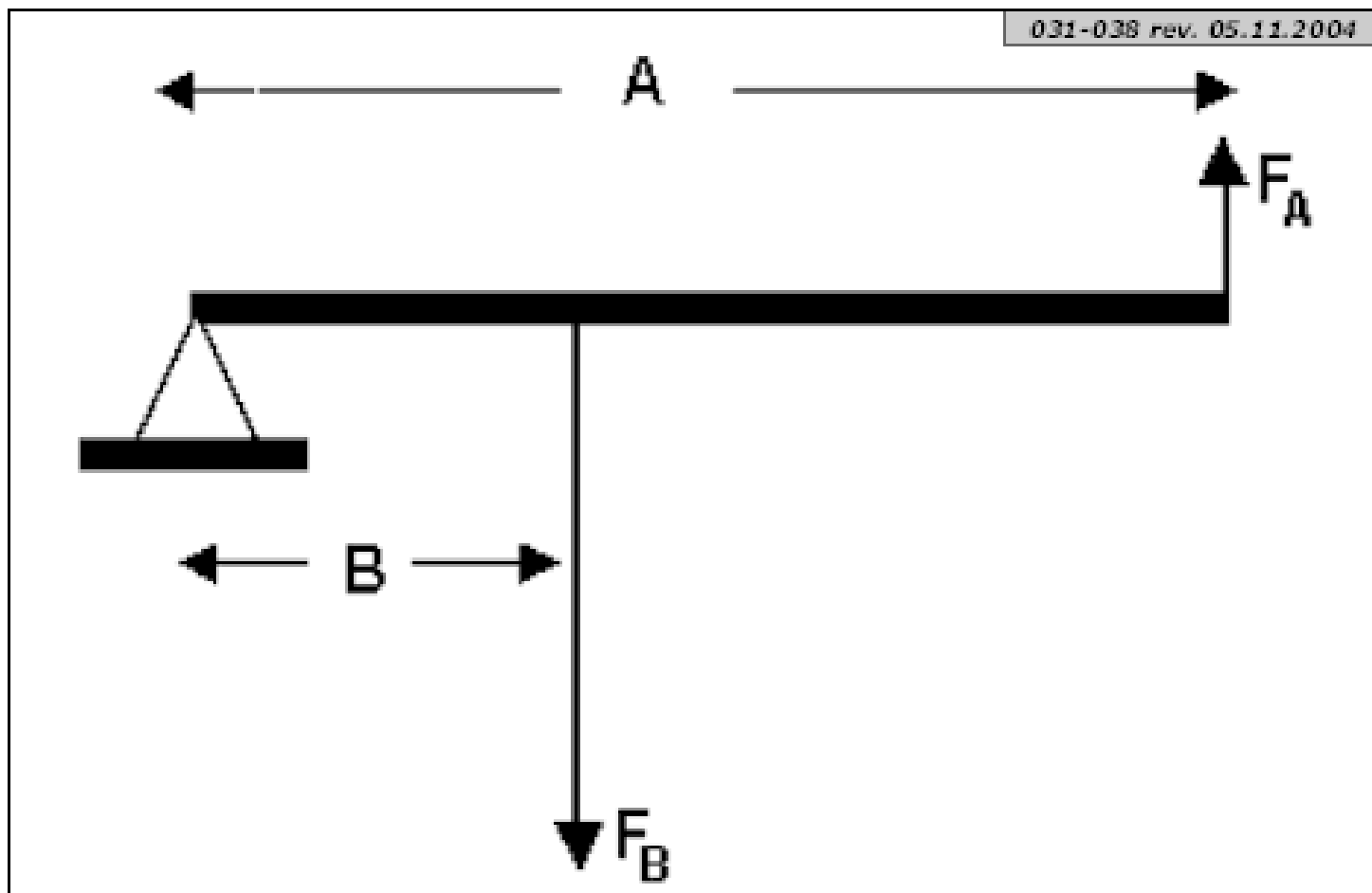
☐ D ZZZZ

## Attachment No. 1

Questions 15

Annex

031-038.gif

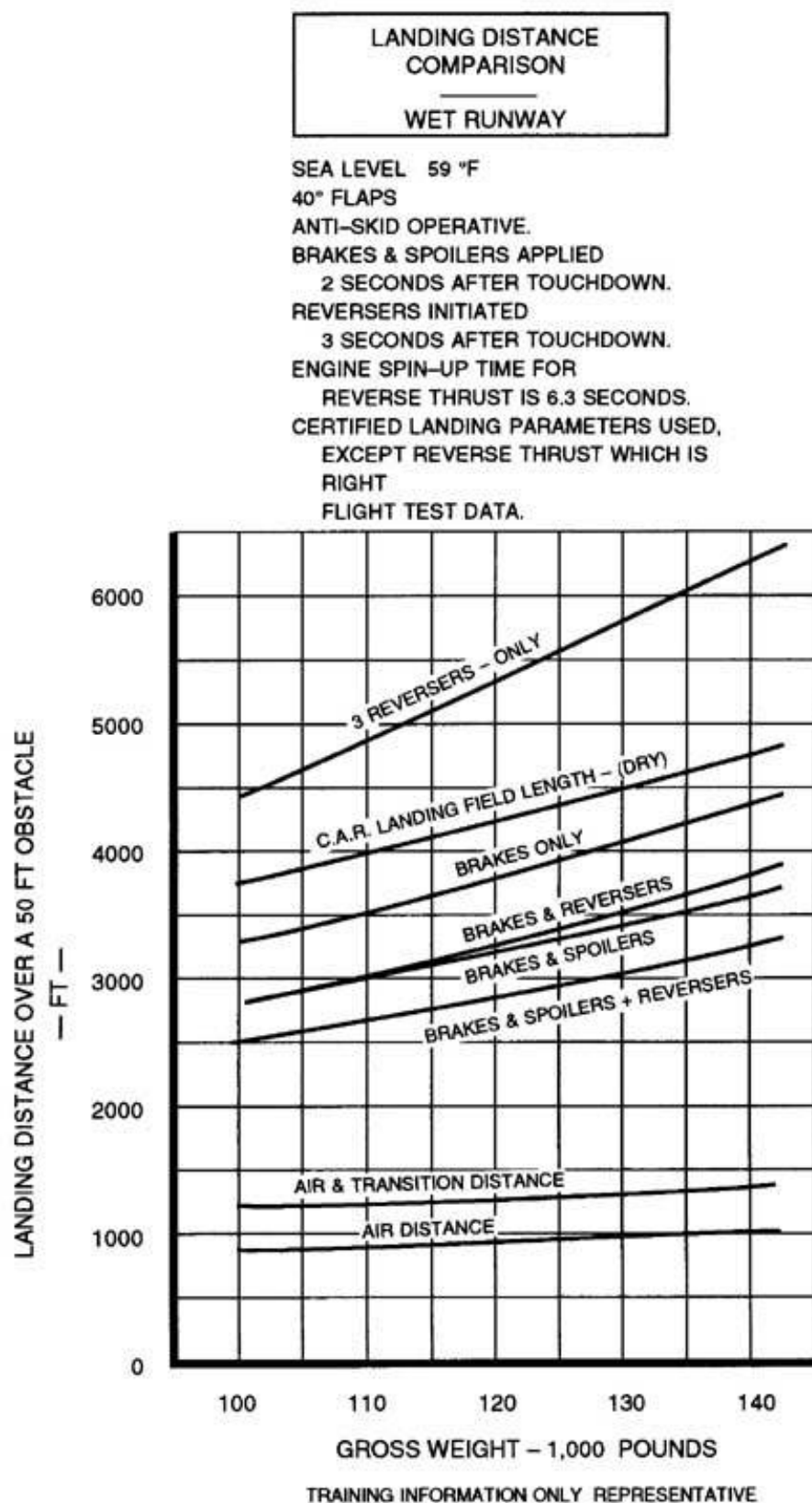


## Attachment No. 2

Questions 17

figure 89

atp\_89.jpg



### NORMAL LANDING

**Figure 89. B-727 – Normal Landing – Wet Runway © ASA**